Korobochka (コロボ) @cirnosad

Nov 1, 2023 • 18 tweets • cirnosad/status/1719731366848578037

PHYS004A: Improving the nuclear bomb, and scaling up!

In this first part of a multi-part thread we are going to learn about the process by which "Gadget" and "Fat man" were turned into efficient, compact and more practical nuclear bombs... with way more yield!

In the previous PHYS threads we discussed:

PHYS001 the principles of fission in depth:

PHYS002 General bomb design:

PHYS003 fission bomb design:

Today we are joined by a guest, Okuu, please be kind to her.

Korobochka (コロボ) 🇦🇺✝️🇷🇺
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⚛️PHYS001: Basics of atomic physics and fission.

In this short thread-course, I'm going to get you up to speed with the basics of atomic physics.
If you already understand the words "chain reaction", "mass defect" and "fission", you can skip this course.
Everyone else, let's go!

1:06 PM · May 18, 2023

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PHYS002: Bomb design/Super-basic thermodynamics

In this short thread, we're going to take a step back from nuclear weapon design to look at conventional bombs. You'll appreciate the engineering challenges facing designers and how they optimize the devices.

Let's go!

6:19 AM · May 20, 2023

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PHYS003: Fission bomb design & history
In this short thread, we're going to learn about "critical mass", timing circuits and with them, we'll combine what we learn in PHYS001 & PHYS002 to learn how the ultimate weapon was created:
The nuclear bomb!
Let's discuss how it works too!

9:57 PM · May 23, 2023

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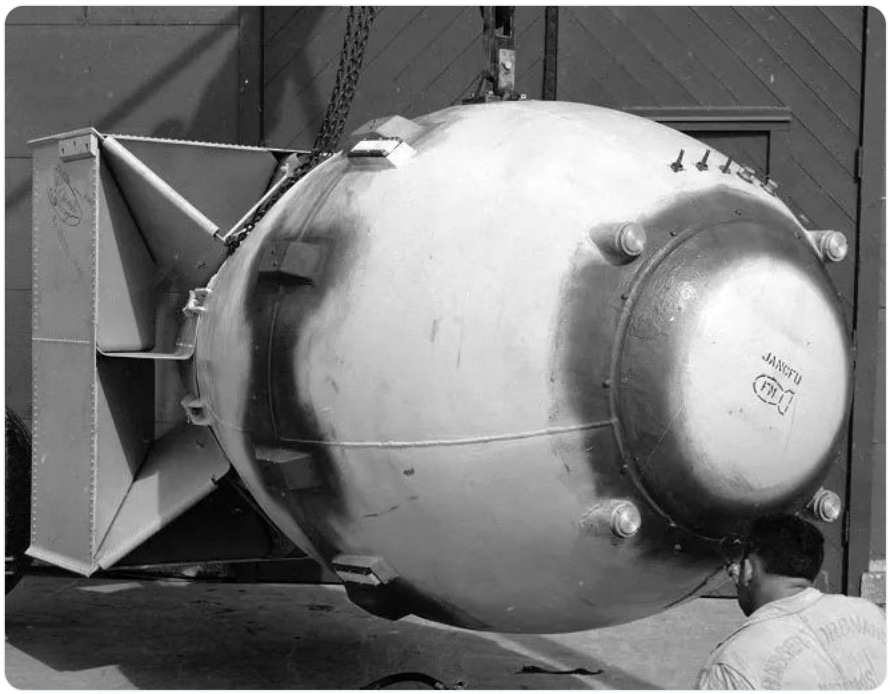
Fat man was large (3.5mx1.5m 4.7ton) with only 15kT yield, generals wanted nukes which were:

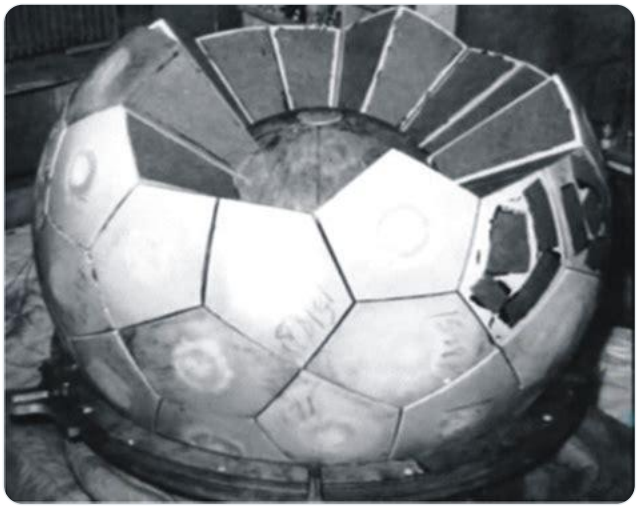
* Cheaper: Fissile materials were very expensive and took a great time to enrich.

* Less heavy: Can be used/delivered more easily.

* More powerful: It was fashionable to threaten cities



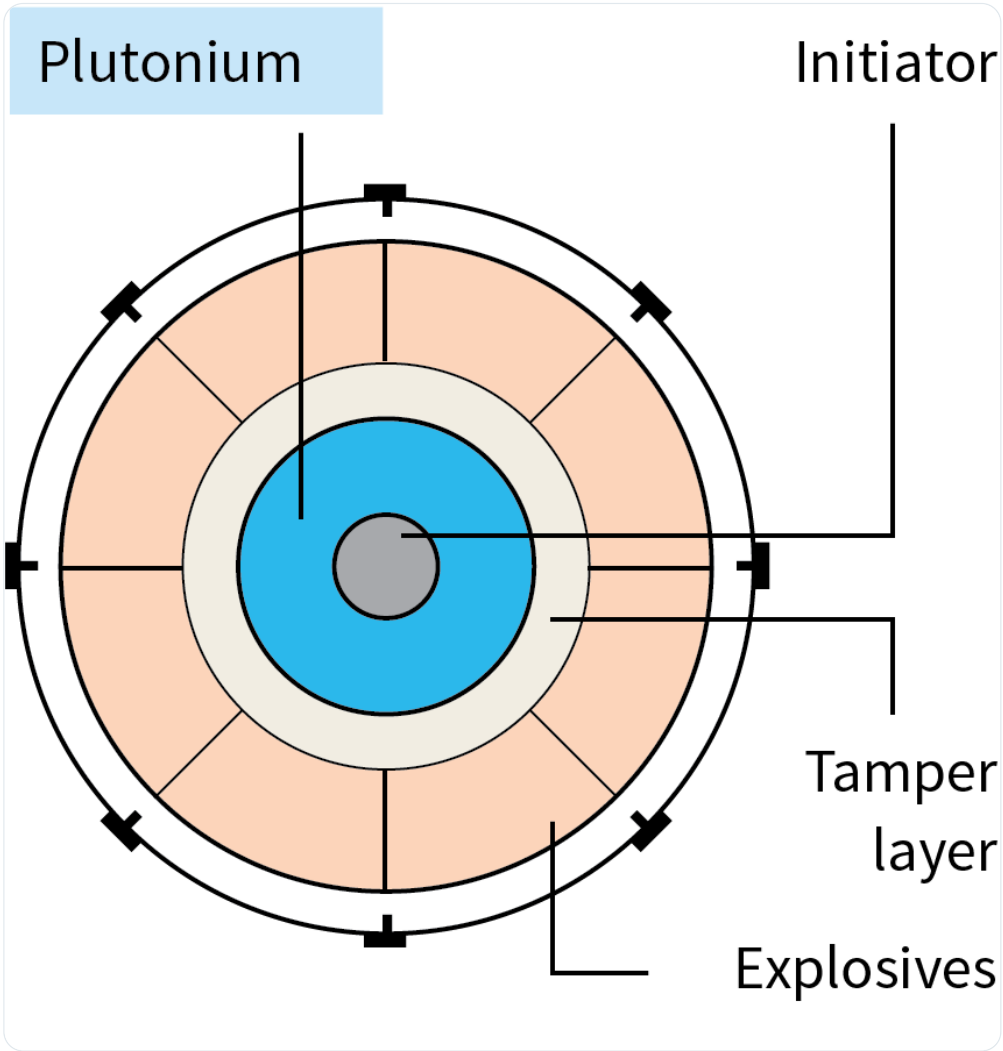
Fat man's implosion design used a 32-point high-explosive lens. It made use of a 6.2kg δ-phase plutonium-gallium (4%) pit for practical reasons. Note that this is below the actual critical mass at standard temperatures of 10kg. But it will still go super critical for 3 reasons.



1. The shape charge will compress the pit dramatically, causing the Plutonium atoms to close in spacing.
2. Plutonium has an a-phase allotrope that is about 25% denser, phase transitioned to during detonation. A surprising bonus!
3. Increasing neutron flux via reflector (tamper).

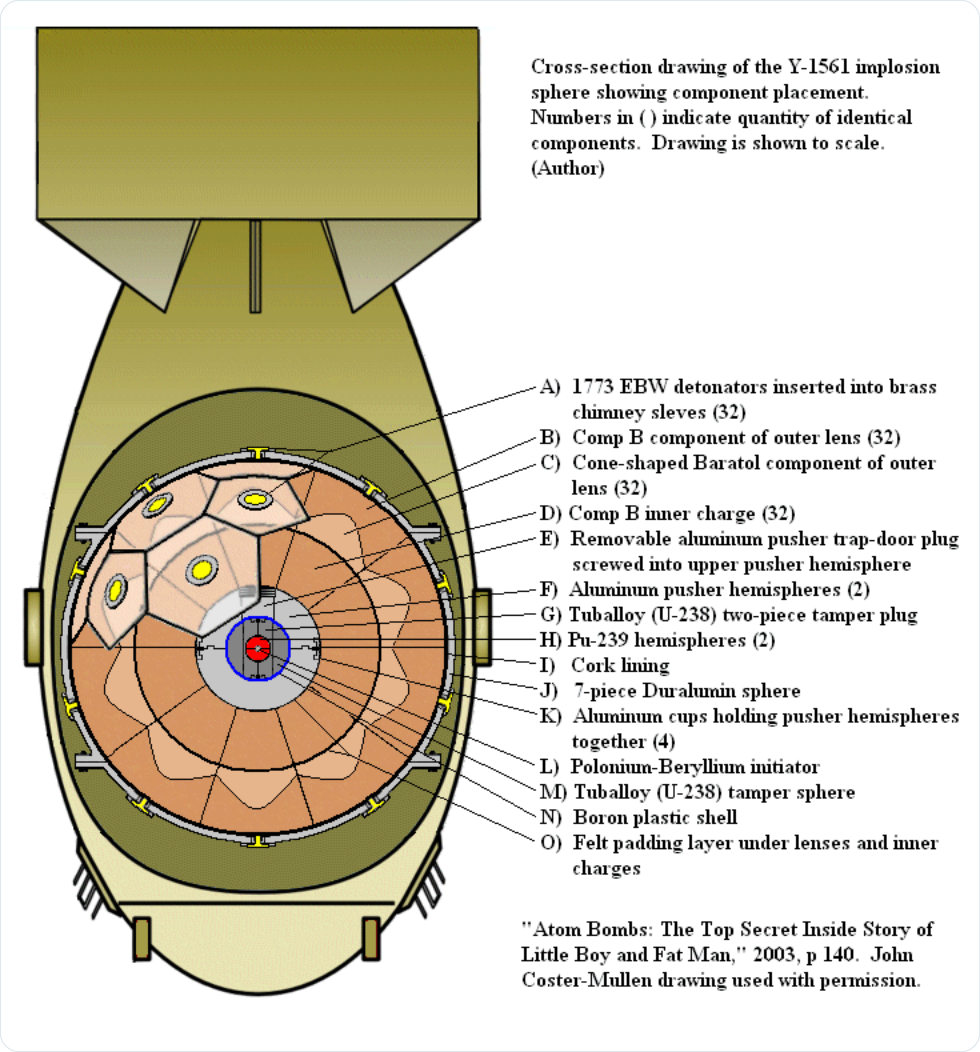
For #3, the pit was placed in a tamper made of U-238 (DU) coated with Beryllium-doped plastic which reflected neutrons.

All in all, only 16% of the Plutonium fissioned! But, the DU *also* underwent fission when exposed to energetic neutrons, accounting for 30% of the yield!



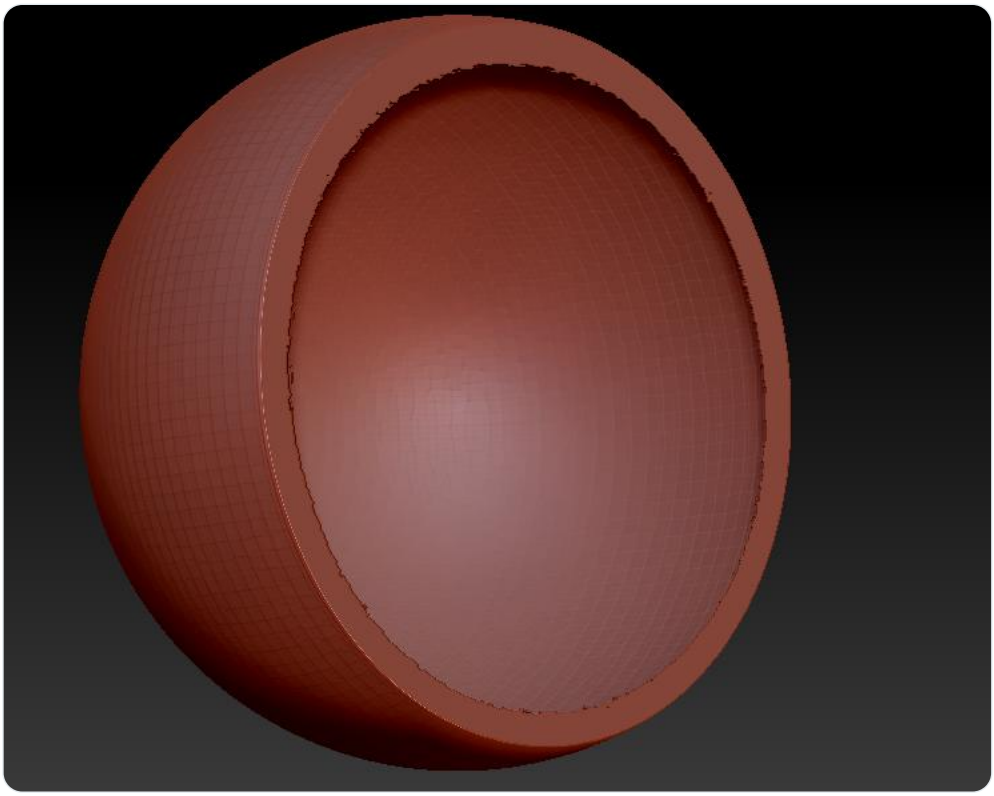
These components were part of the "physics package" & accounted for a small % of the bomb's weight.

The majority of the mass was in the high explosives and the bomb casing. Recall, from PHYS001, that the latter is needed to hold the bomb together until the reactions "complete".

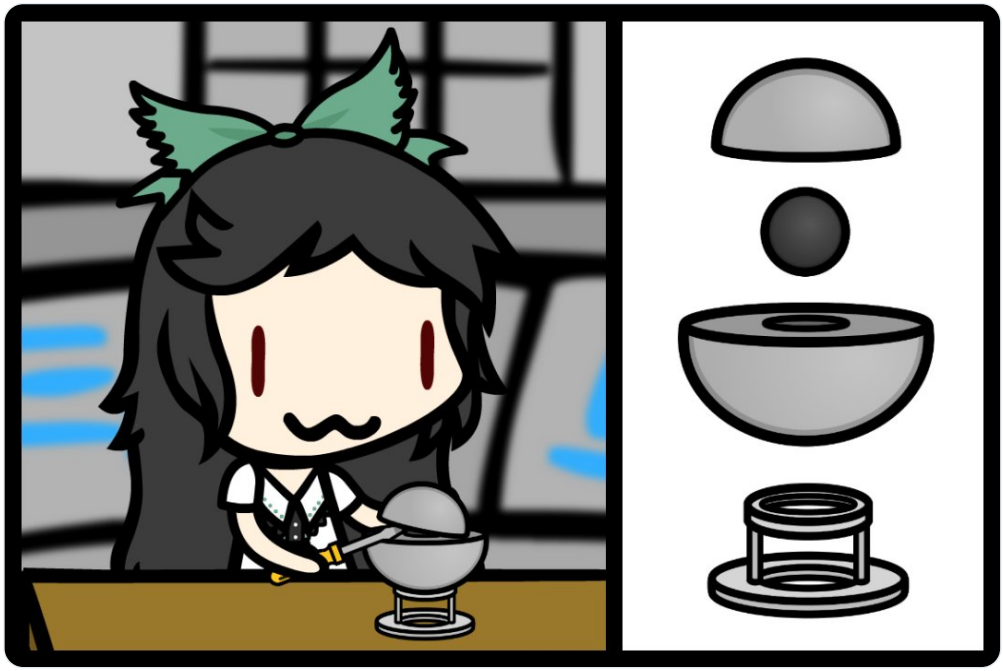


There was a room for improvements!

The pit was a solid sphere and this wasn't ideal! If it were a hollowed out sphere, the compression wave would crush it inwards into the neutron source (initiator) with huge deformation. This has a highly non-linear advantage in criticality!



The hollow sphere design also self-tampered, meaning the deformation formed a neutron reflector out of the plutonium itself, and a highly super-critical region in the centre. Though this required more precise shaped charge, it required less explosives and a lighter casing!

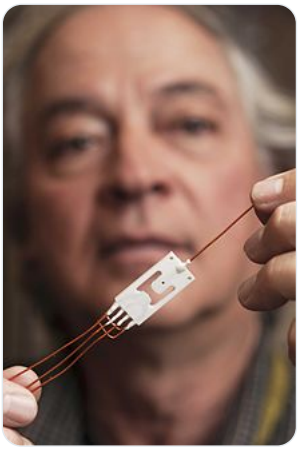


Another improvement was the levitated pit, essentially allowing an air gap to exist between the tamper and the pit. This created a shockwave that increased the compression even more, resulting in twice the yield.

(Photo for illustrative purposes only. A cone was said to be used)

The initiator (usually polonium) could be replaced with a pulsed neutron emitter to avoid pre-detonation, which would reduce the yield. Eventually this would be integrated into a chip!

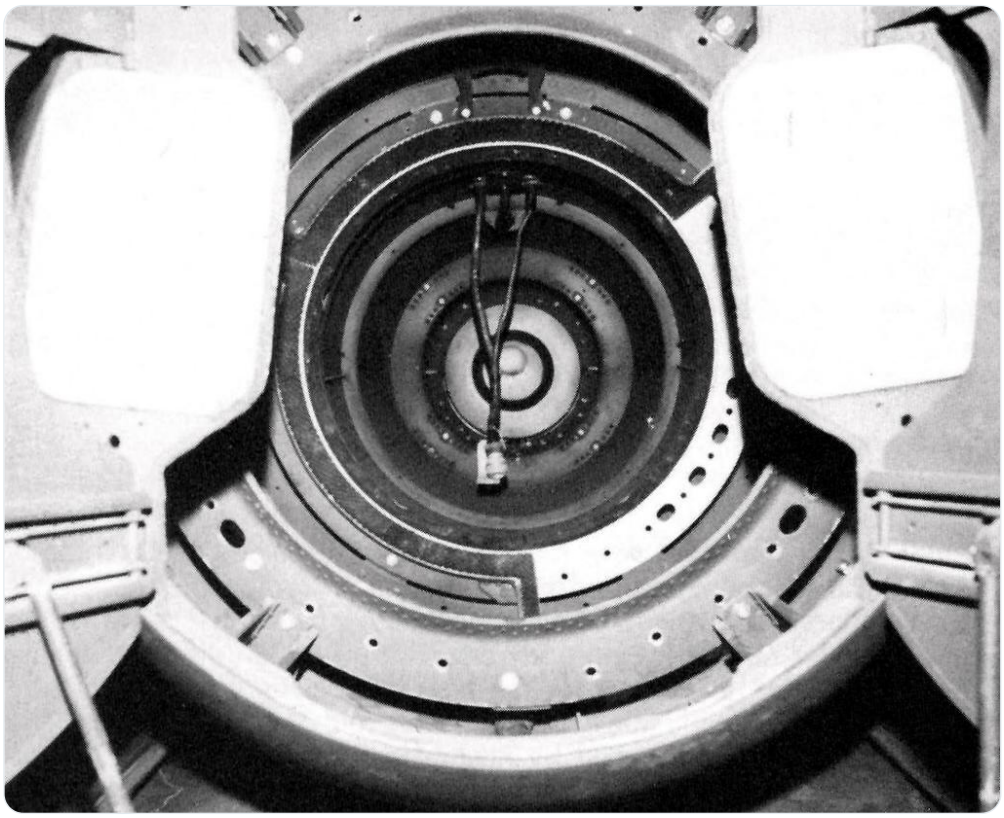
Finally, a more precise 92-point shape charge would enable the use of the hollow pit.



These improvements lead to two paths.

* Highly efficient bombs such as the Mark 5, with a yield up-to 120kT, burning up most of the (Plutonium-Uranium) pit and making good use of the tamper, while weighting only 1.2 tons.

* Light bombs, the Mark 12 weighing 540kg with yield 14kT.



But for some, 120 kilotons (almost 6 times Nagasaki) wasn't enough!

The Mk 18 achieved a yield of 500kT. It was quite similar to the Mk 5 design but employed an unsafe 60kg HEU pit, which had Aluminium/Boron reflectors inserted to stop it from pre-detonating on its own.

This was of course a dead end, as there was no way to scale the design further due to the safety concern -- the reflectors were removed at the last step of arming.

But that didn't stop the UK from doing something almost equally insane! Orange Herald had a yield of 720kT!

Orange Herald wasted 117kg of HEU (almost the annual production of the UK at the time, ugh! 😟), and used a hollow pit with a special* material inside. The material did not operate as expected and the device was considered a failure.

* We'll get to what it was in PHYS004B!

Fission based nukes seemed to reach the end of the line here, you could not really improve upon them without waiting for better materials and electronics.

But for one mad scientist, these bombs were merely a stepping stone to something bigger: The super bomb, aka a fusion bomb!

... But before we can cover what the "Super" was meant to and what came of it, we will need to revise three subjects:

* Thermodynamics
* Atomic physics from a coulomb law perspective and open questions regarding the structure of the atom/subatomic particles.
* Nuclear fusion! 😄

Thank you for reading! Join me and Okuu next time for PHYS004B.

And, sorry I couldn't fit it all in a single thread, but the concepts discussed so far are important enough to require your close attention. As homework, think about what advantages a hollow pit could bring!

/End

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